

Basic Disk Management in Linux (Partitioning & Mounting a New Disk)

Disk Management

- Think of your hard drive as a big empty room. Disk Management helps divide it into smaller "rooms" (partitions or volumes) for different purposes, like one for Windows, one for files, or one for backups. You access it by right-clicking "This PC" > Manage > Disk Management. It shows all your drives, their sizes, and status.

Primary Partition Explained

- A primary partition is a main section on the disk that can store files, programs, or even the Windows operating system. On older MBR-style disks (common in legacy systems), you can have up to four primary partitions. One of them is often marked "active" to boot your computer, like the C: drive where Windows lives. It's simple and straightforward for basic setups.

Extended Partition Explained

- An extended partition is like a special container that takes up one of your four primary slots on an MBR disk. Inside it, you create multiple "logical drives" (extra sections with drive letters like D:, E:). This workaround lets you have more than four total sections without switching to modern GPT disks. The extended partition itself doesn't get a drive letter—only the logical drives inside do.

How to create/manage partitions in a new disk

Step 1: Check for New Disks with fdisk -l

- Run `sudo fdisk -l` as root to list all disks and partitions. This scans the system for block devices like `/dev/nvme0n1` (NVMe SSDs) or `/dev/sda` (SATA). Look for your new disk—e.g., `/dev/nvme0n1` with no partitions listed under it. Use `lsblk` for a tree view if needed.

Step 2: Enter Interactive fdisk Mode

- Run `sudo fdisk /dev/nvme0n1` (replace "nvme0n1" with your disk, like nvme0n2). You'll enter interactive mode with a prompt like "Command (m for help):". Type `m` to see all commands, `p` to print current table (empty for new disk).

Step 3: Create Partition in Interactive Mode

- Type `g` to create a new GPT partition table (modern, no primary/extended limits; skip for MBR with `o`).
- `n` for new partition: Select partition number (default 1), first sector (default), last sector (e.g., `+100G` or full disk with default).
- `t` to set type: Choose 83 for Linux (or `bf` for XFS if prompted), `w` to write changes and exit.
- Example session: `g, n, enter (defaults), Enter, w.`

Using partprobe

- `sudo partprobe /dev/nvme0n1` (or just `sudo partprobe`) informs the kernel of partition table changes without rebooting. It re-reads disks so new partitions like `/dev/nvme0n1p1` appear in `lsblk` or `fdisk -l` immediately.

Formatting with mkfs.xfs'

- `sudo mkfs.xfs /dev/nvme0n1p1` formats the new partition (p1 = partition 1) as XFS filesystem (RHEL default for high-performance storage). Add -f to force if needed. XFS supports large files/drives; verify with `blkid /dev/nvme0n1p1` (shows TYPE="xfs")

Mounting via /etc/fstab (Permanent)

- Create mount point: `sudo mkdir /mnt/data`.
- Get UUID: `blkid /dev/nvme0n1p1` (e.g., UUID="abc-123").
- Edit fstab: `sudo vi /etc/fstab`, add line: UUID=abc-123 /mnt/data xfs defaults 0 0.
- Test: `sudo mount -a` (mounts all fstab without reboot), `df -h` to verify.
- Reboot to confirm persistence. Backup fstab first!

To delete the partition

- Enter interactive mode: `fdisk /dev/nvme0n1`
- d for deleting partition: select the partition number to be deleted
- w to write -> save the changes